

Is there a noncommutative spectrum functor?

Manuel Reyes

Spectrums in Representation Theory of Algebras conference

December 18, 2025

University of California, Irvine

Noncommutative spectral theory

The spectrum

A **spectrum** is an assignment

$$\{\text{commutative algebras}\} \rightarrow \{\text{spaces}\}$$

Example: For a commutative ring R , its **Zariski spectrum** is

$$\text{Spec}(R) = \{\text{prime ideals of } R\}.$$

It carries a *Zariski topology*, and is nonempty if $R \neq 0$ (by Zorn's lemma).

The spectrum

A **spectrum** is an assignment

$$\{\text{commutative algebras}\} \rightarrow \{\text{spaces}\}$$

Example: For a commutative ring R , its **Zariski spectrum** is

$$\text{Spec}(R) = \{\text{prime ideals of } R\}.$$

It carries a *Zariski topology*, and is nonempty if $R \neq 0$ (by Zorn's lemma).

In algebraic geometry it is equipped with a *structure sheaf* to produce a *scheme*, whose ring of global sections is

$$\Gamma(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)}) \cong R.$$

Further examples of spectra

Ex. 2: For a (unital) commutative C^* -algebra A , its **Gelfand spectrum** is

$$\mathrm{Spec}(A) = \{\text{maximal ideals of } A\}.$$

This is a *compact Hausdorff space*, and we may recover A from this spectrum as $C(\mathrm{Spec}(A)) \cong A$.

Further examples of spectra

Ex. 2: For a (unital) commutative C^* -algebra A , its **Gelfand spectrum** is

$$\mathrm{Spec}(A) = \{\text{maximal ideals of } A\}.$$

This is a *compact Hausdorff space*, and we may recover A from this spectrum as $C(\mathrm{Spec}(A)) \cong A$.

Ex. 3: For a Boolean algebra B , its **Stone spectrum** is

$$\mathrm{Spec}(B) = \{\text{prime ideals of } B\} \cong \{\text{ultrafilters of } B\},$$

where $P \subseteq B$ is an *ideal* if it is non-empty, $\vee$ -closed down-set, and it is *prime* if $a \wedge b \in P \implies a \in P$ or $b \in P$. This is a compact, 0-dimensional (= *Stone*) space, from which we can recover B as the lattice of clopen sets.

Spectrum as a link to noncommutative geometry

Thus we have:

- Commutative rings: $\text{Spec}(R) = \{\text{prime ideals of } R\}$
- Commutative C^* -algebras: $\text{Spec}(A) = \{\text{max. ideals of } A\}$
- Boolean algebra: $\text{Spec}(B) = \{\text{prime ideals of } B\}$

Each of these spectra is a (contravariant) **functor**: each algebra morphism $f: A \rightarrow B$ yields a map between spaces $\text{Spec}(B) \rightarrow \text{Spec}(A)$ by $P \mapsto f^{-1}(P)$.

Spectrum as a link to noncommutative geometry

Thus we have:

- Commutative rings: $\text{Spec}(R) = \{\text{prime ideals of } R\}$
- Commutative C*-algebras: $\text{Spec}(A) = \{\text{max. ideals of } A\}$
- Boolean algebra: $\text{Spec}(B) = \{\text{prime ideals of } B\}$

Each of these spectra is a (contravariant) **functor**: each algebra morphism $f: A \rightarrow B$ yields a map between spaces $\text{Spec}(B) \rightarrow \text{Spec}(A)$ by $P \mapsto f^{-1}(P)$.

This picture is often given as the motivation behind **noncommutative geometry** of various flavors:

$$\{\text{noncommutative algebras}\} \longleftrightarrow \{\text{“noncommutative spaces”}\}$$

Motivating question

Noncommutative spectral theory: Creating and studying invariants of NC rings that extend the usual spectrum of a commutative ring or algebra

Includes many well-loved subjects: Prime ideal spectra, prime left ideals, primitive ideals, state spaces, localization theories, spectra of abelian/triangulated categories...

Motivating question

Noncommutative spectral theory: Creating and studying invariants of NC rings that extend the usual spectrum of a commutative ring or algebra

Includes many well-loved subjects: Prime ideal spectra, prime left ideals, primitive ideals, state spaces, localization theories, spectra of abelian/triangulated categories...

In some sense, these try address:

Question: What is the “noncommutative space” corresponding to a noncommutative algebra?

Functorial spectral theory

Functorial spectral theory: Searching for an invariant that extends Spec as a **functor** from commutative to NC rings

Why functorial?

- Maintains “algebra-geometry” correspondence.
- Guarantees that we can try to understand spectrum of an algebra by looking at related subalgebras.

Functorial spectral theory

Functorial spectral theory: Searching for an invariant that extends Spec as a **functor** from commutative to NC rings

Why functorial?

- Maintains “algebra-geometry” correspondence.
- Guarantees that we can try to understand spectrum of an algebra by looking at related subalgebras.

Familiar NC spectra tend not to be functorial. The few functorial ones tend to trivialize often. Sadly, this is not an accident...

Obstructions to Spec functors

Note: If we even hope to build a space from a ring, we first need an underlying set. So we'll ask about functors to **Set**.

Obstructions to Spec functors

Note: If we even hope to build a space from a ring, we first need an underlying set. So we'll ask about functors to **Set**.

Theorem (R., 2012)

Any functor $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ whose restriction to the full subcategory $\mathbf{cRing}^{\text{op}}$ is isomorphic to $\mathbf{Spec}$ has $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for $n \geq 3$.

Similarly, any functor $F: \mathbf{Cstar}^{\text{op}} \rightarrow \mathbf{Set}$ whose restriction to the full subcategory $\mathbf{cCstar}^{\text{op}}$ is isomorphic to $\mathbf{Spec}$ has $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for $n \geq 3$.

Idea of the proof: three steps

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 1: Find a **universal** example of such $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$.

This is a functor $p\text{-Spec}: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ such that there are unique *natural* maps

$$F(R) \rightarrow p\text{-Spec}(R)$$

for any possible F and all rings R .

Idea of the proof: three steps

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 1: Find a **universal** example of such $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$.

This is a functor $p\text{-Spec}: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ such that there are unique *natural* maps

$$F(R) \rightarrow p\text{-Spec}(R)$$

for any possible F and all rings R . Concretely,

$$p\text{-Spec}(R) = \{Q \subseteq R \mid Q \cap C \in \mathbf{Spec}(C) \text{ for all comm. } C \subseteq R\}$$

Idea of the proof

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 2: Show that this universal example assigns $\mathbb{M}_n(\mathbb{C}) \mapsto \emptyset$.

Idea: For a given $Q \in p\text{-Spec}(\mathbb{M}_n(\mathbb{C}))$, we can “color” a given projection $p \in \mathbb{M}_n(\mathbb{C})$:

$$p \mapsto \begin{cases} 0, & p \in Q \\ 1, & p \notin Q \end{cases}$$

Fact: If $I_n = p_1 + \cdots + p_n$, then one $p_i \mapsto 1$, the rest are 0.

Idea of the proof

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 2: Show that this universal example assigns $\mathbb{M}_n(\mathbb{C}) \mapsto \emptyset$.

Idea: For a given $Q \in p\text{-Spec}(\mathbb{M}_n(\mathbb{C}))$, we can “color” a given projection $p \in \mathbb{M}_n(\mathbb{C})$:

$$p \mapsto \begin{cases} 0, & p \in Q \\ 1, & p \notin Q \end{cases}$$

Fact: If $I_n = p_1 + \cdots + p_n$, then one $p_i \mapsto 1$, the rest are 0.

A surprise: This is a **Kochen-Specker** coloring, which shows up in the logical foundations of Quantum Mechanics!

The Kochen-Specker Theorem

Hidden variables problem (roughly): Can all observables be simultaneously given definite values, which are independent of the device used to measure them?

- Observables: self-adjoint matrices $\mathbb{M}_n(\mathbb{C})_{sa}$
- Definite values: function $f: \mathbb{M}_n(\mathbb{C})_{sa} \rightarrow \mathbb{R}$
- “Yes-No” observable: projection $p = p^2 = p^* \in \mathbb{M}_n(\mathbb{C})$, values $\{0, 1\}$

The Kochen-Specker Theorem

Hidden variables problem (roughly): Can all observables be simultaneously given definite values, which are independent of the device used to measure them?

- Observables: self-adjoint matrices $\mathbb{M}_n(\mathbb{C})_{sa}$
- Definite values: function $f: \mathbb{M}_n(\mathbb{C})_{sa} \rightarrow \mathbb{R}$
- “Yes-No” observable: projection $p = p^2 = p^* \in \mathbb{M}_n(\mathbb{C})$, values $\{0, 1\}$

Such f restricts to a Kochen-Specker coloring on projections.

Equivalently: f is “Boolean whenever there is no uncertainty.”

The Kochen-Specker Theorem

Hidden variables problem (roughly): Can all observables be simultaneously given definite values, which are independent of the device used to measure them?

- Observables: self-adjoint matrices $\mathbb{M}_n(\mathbb{C})_{sa}$
- Definite values: function $f: \mathbb{M}_n(\mathbb{C})_{sa} \rightarrow \mathbb{R}$
- “Yes-No” observable: projection $p = p^2 = p^* \in \mathbb{M}_n(\mathbb{C})$, values $\{0, 1\}$

Such f restricts to a Kochen-Specker coloring on projections.

Equivalently: f is “Boolean whenever there is no uncertainty.”

Kochen-Specker Theorem (1967)

There is no Kochen-Specker coloring of $\text{Proj}(\mathbb{M}_n(\mathbb{C}))$ for $n \geq 3$.

Corollary: $p\text{-Spec}(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for $n \geq 3$.

Idea of the proof

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 3: By universality, conclude that *every* F does the same.

Proof: Denote $R = \mathbb{M}_n(\mathbb{C})$, $n \geq 3$. Kochen-Specker implies that $p\text{-Spec}(R) = \emptyset$. By universality, for any such F we have a function

$$F(R) \rightarrow p\text{-Spec}(R) = \emptyset.$$

Idea of the proof

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 3: By universality, conclude that *every* F does the same.

Proof: Denote $R = \mathbb{M}_n(\mathbb{C})$, $n \geq 3$. Kochen-Specker implies that $p\text{-Spec}(R) = \emptyset$. By universality, for any such F we have a function

$$F(R) \rightarrow p\text{-Spec}(R) = \emptyset.$$

The only set with a map to the empty set is the empty set,

Idea of the proof

Theorem (R., 2012):

Let $F: \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Set}$ be any functor whose restriction to commutative rings is $\mathbf{Spec}$. Then $F(\mathbb{M}_n(\mathbb{C})) = \emptyset$ for all $n \geq 3$.

Step 3: By universality, conclude that *every* F does the same.

Proof: Denote $R = \mathbb{M}_n(\mathbb{C})$, $n \geq 3$. Kochen-Specker implies that $p\text{-Spec}(R) = \emptyset$. By universality, for any such F we have a function

$$F(R) \rightarrow p\text{-Spec}(R) = \emptyset.$$

The only set with a map to the empty set is the empty set, so $F(R) = \emptyset$ as well! □

Some loose ends

Q1: What happens for $n = 2$?

Q2: What is so special about $\mathbb{C}$?

Some loose ends

Q1: What happens for $n = 2$?

A: The functor $F = p\text{-Spec}$ sends $\mathbb{M}_2(\mathbb{C})$ to a set of cardinality $2^{\mathfrak{c}} = 2^{2^{\aleph_0}}$. It's huge!

Q2: What is so special about $\mathbb{C}$?

Some loose ends

Q1: What happens for $n = 2$?

A: The functor $F = p\text{-Spec}$ sends $\mathbb{M}_2(\mathbb{C})$ to a set of cardinality $2^{\mathfrak{c}} = 2^{2^{\aleph_0}}$. It's huge!

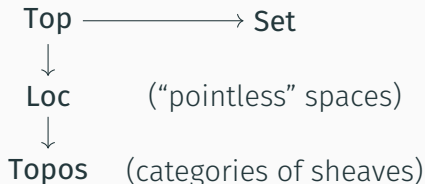
Q2: What is so special about $\mathbb{C}$?

A: It was the original context for the Kochen-Specker Theorem. The obstruction still holds with $\mathbb{C}$ replaced by any complex algebra (e.g., any C^* -algebra).

Later work (Ben-Zvi, Ma, R. 2017) found an “integer-valued” version of Kochen-Specker, so $\mathbb{C}$ can be replaced by *any* ring.

Topology without sets

Does the problem lie with points? There are “point-free” approaches to topology. Try a different forgetful functor?



Theorem (van den Berg, Heunen 2014): The same type of obstruction holds when viewing **Spec** as a functor to **Loc** or **Topos**.

Topological methods in categories

Algebraic geometers have other abstract ways to do geometry...

Grothendieck topologies: The category $\mathcal{C} = \mathbf{cRing}^{\mathrm{op}}$ can be equipped with a certain Grothendieck topology, producing the **big Zariski site**. Every commutative ring is a sheaf for this topology.

Theorem (R., 2014)

We cannot extend this to a Grothendieck topology on $\mathbf{Ring}^{\mathrm{op}}$ in such a way that matrix rings $\mathbb{M}_n(R)$ ($n \geq 2$) are sheaves for the topology.

My interpretation: Sets and topologies are too “classical” to capture the “quantum” nature of noncommutative geometry.

It would be more appropriate to replace classical sets with **noncommutative sets**, i.e. noncommutative **discrete** spaces.

Where possible, we should try to incorporate lessons from quantum information theory!

A fundamental question

Q: What objects should be viewed as *noncommutative sets* (i.e., NC *discrete* spaces) in NCG?

Ideal solution: a fully faithful embedding $\mathbf{Set} \hookrightarrow \mathfrak{G}$ into a category of NC sets, and a NC spectrum functor $\Sigma: \mathbf{Ring}^{\text{op}} \rightarrow \mathfrak{G}$ with commuting diagram

$$\begin{array}{ccccc} \mathbf{cRing}^{\text{op}} & \xrightarrow{\text{Spec}} & \mathbf{Top} & \xrightarrow{U} & \mathbf{Set} \\ \downarrow & & & & \downarrow \\ \mathbf{Ring}^{\text{op}} & \xrightarrow{\Sigma} & & & \mathfrak{G} \end{array}$$

Compromises for now: Replace $\text{Spec} \rightsquigarrow \text{Max}$, restrict the class of algebras, and find an approximate solution...

Quantizing the maximal spectrum

From sets to “quantum sets”

$$\begin{array}{ccc} \{\text{commutative algebras}\} & \xrightarrow{\text{Spec}} & \{\text{sets}\} \\ \downarrow & & \downarrow \\ \{\text{noncommutative algebras}\} & \longrightarrow & \{\text{???\} \end{array}$$

Inspiration from QM: If X is our set of “states,” we should also allow **linear combinations** of states: $X \rightsquigarrow kX = \text{Span}(X)$

From sets to “quantum sets”

$$\begin{array}{ccc} \{\text{commutative algebras}\} & \xrightarrow{\text{Spec}} & \{\text{sets}\} \\ \downarrow & & \downarrow \\ \{\text{noncommutative algebras}\} & \longrightarrow & \{\text{???\} \end{array}$$

Inspiration from QM: If X is our set of “states,” we should also allow **linear combinations** of states: $X \rightsquigarrow kX = \text{Span}(X)$

This vector space $Q = kX$ carries the structure of a **coalgebra**:

- Comultiplication $\Delta: Q \rightarrow Q \otimes Q$ given by $x \mapsto x \otimes x$
- Counit $\eta: Q \rightarrow k$ given by $x \mapsto 1$

From sets to “quantum sets”

$$\begin{array}{ccc} \{\text{commutative algebras}\} & \xrightarrow{\text{Spec}} & \{\text{sets}\} \\ \downarrow & & \downarrow \\ \{\text{noncommutative algebras}\} & \longrightarrow & \{\text{???\} \end{array}$$

Inspiration from QM: If X is our set of “states,” we should also allow **linear combinations** of states: $X \rightsquigarrow kX = \text{Span}(X)$

This vector space $Q = kX$ carries the structure of a **coalgebra**:

- Comultiplication $\Delta: Q \rightarrow Q \otimes Q$ given by $x \mapsto x \otimes x$
- Counit $\eta: Q \rightarrow k$ given by $x \mapsto 1$

We get “the same” maps: $\mathbf{Set}(X, Y) \cong \mathbf{Coalg}(kX, kY)$. Gives a full and faithful embedding $\mathbf{Set} \hookrightarrow \mathbf{Coalg}$.

“Quantum sets” for algebras over a field

Therefore: We view a coalgebra (Q, Δ, η) as a “quantum set” (over k) with *algebra of observables* $\text{Obs}(Q) = Q^*$.

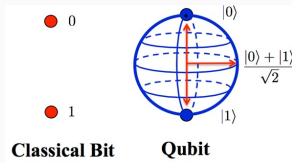
History: Coalgebras were considered as “discrete objects” by Takeuchi (1974), and in the noncommutative context by Kontsevich-Soibelman (*NC thin schemes*) and Le Bruyn.

“Quantum sets” for algebras over a field

Therefore: We view a coalgebra (Q, Δ, η) as a “quantum set” (over k) with *algebra of observables* $\text{Obs}(Q) = Q^*$.

History: Coalgebras were considered as “discrete objects” by Takeuchi (1974), and in the noncommutative context by Kontsevich-Soibelman (*NC thin schemes*) and Le Bruyn.

Ex: $\mathbb{M}^d = \mathbb{M}_d(k)^*$ has “algebra of observables” $(\mathbb{M}^d)^* \cong \mathbb{M}_d(k)$. Thus we may view it as a quantum d -level system or **qudit**.



The quantized maximal spectrum

Observations:

1. For $k = \bar{k}$ and A commutative affine, $\text{Max}(A) \cong \mathbf{Alg}(A, k)$.

The quantized maximal spectrum

Observations:

1. For $k = \bar{k}$ and A commutative affine, $\text{Max}(A) \cong \mathbf{Alg}(A, k)$.
2. The category **Alg** is enriched over **Coalg** using the *measuring coalgebra* construction: $[A, B] = P(A, B)$.

The quantized maximal spectrum

Observations:

1. For $k = \bar{k}$ and A commutative affine, $\text{Max}(A) \cong \mathbf{Alg}(A, k)$.
2. The category **Alg** is enriched over **Coalg** using the *measuring coalgebra* construction: $[A, B] = P(A, B)$.
3. Batchelor viewed this as a “quantized set of maps” between the noncommutative spaces.

The quantized maximal spectrum

Observations:

1. For $k = \bar{k}$ and A commutative affine, $\text{Max}(A) \cong \mathbf{Alg}(A, k)$.
2. The category $\mathbf{Alg}$ is enriched over $\mathbf{Coalg}$ using the *measuring coalgebra* construction: $[A, B] = P(A, B)$.
3. Batchelor viewed this as a “quantized set of maps” between the noncommutative spaces.
4. The “quantized set of maps” from A to k is the **finite dual coalgebra**:

$$A^\circ := \varinjlim_{I \text{ fin. codim.}} (A/I)^* \cong [A, k]$$

The quantized maximal spectrum

Observations:

1. For $k = \bar{k}$ and A commutative affine, $\text{Max}(A) \cong \mathbf{Alg}(A, k)$.
2. The category $\mathbf{Alg}$ is enriched over $\mathbf{Coalg}$ using the *measuring coalgebra* construction: $[A, B] = P(A, B)$.
3. Batchelor viewed this as a “quantized set of maps” between the noncommutative spaces.
4. The “quantized set of maps” from A to k is the **finite dual coalgebra**:

$$A^\circ := \varinjlim_{I \text{ fin. codim.}} (A/I)^* \cong [A, k]$$

Thesis: For affine k -algebras with “many” f.d. rep’s, the functor $A \mapsto A^\circ$ is a suitable candidate for a **quantized maximal spectrum**.

“Underlying coalgebras” of k -schemes

How does A° compare to $\text{Max}(A)$ in the commutative case? One way to compare is assigning coalgebras to schemes.

Def: For a k -scheme X , the **coalgebra of distributions** is

$$\text{Dist}(X) = \varinjlim \Gamma(S, \mathcal{O}_S)^*,$$

where S ranges over the closed subschemes of X that are finite over k . This gives a functor $\text{Dist}: \mathbf{Sch}_k \rightarrow \mathbf{Coalg}$.

“Underlying coalgebras” of k -schemes

How does A° compare to $\text{Max}(A)$ in the commutative case? One way to compare is assigning coalgebras to schemes.

Def: For a k -scheme X , the **coalgebra of distributions** is

$$\text{Dist}(X) = \varinjlim \Gamma(S, \mathcal{O}_S)^*,$$

where S ranges over the closed subschemes of X that are finite over k . This gives a functor $\text{Dist}: \mathbf{Sch}_k \rightarrow \mathbf{Coalg}$.

Facts: If X is finite type over k , with closed points X_0 :

1. $\text{Dist}(X) \cong \bigoplus_{x \in X_0} (\mathcal{O}_{X,x})^\circ$ as coalgebras.
2. If $k = \bar{k}$, then $kX_0 \hookrightarrow \text{Dist}(X)$

Moral: $\text{Dist}(X)$ **linearizes** the set of closed points, and includes the **formal neighborhood** of each point.

A necessary evil

Affine case: For a commutative affine k -algebra A ,

$$A^\circ \cong \text{Dist}(\text{Spec } A) \cong \bigoplus_{\mathfrak{m} \in \text{Max}(A)} (A_{\mathfrak{m}})^\circ$$

We end up with a diagram similar to our goal:

$$\begin{array}{ccccc} \mathbf{cAff}^{\text{op}} & \xrightarrow{\text{Max}} & \mathbf{Top} & \xrightarrow{U} & \mathbf{Set} \\ \downarrow & & & & \downarrow \\ \mathbf{Alg}^{\text{op}} & \xrightarrow{(-)^\circ} & & & \mathbf{Coalg} \end{array}$$

But it **does not** commute!

A necessary evil

Affine case: For a commutative affine k -algebra A ,

$$A^\circ \cong \text{Dist}(\text{Spec } A) \cong \bigoplus_{\mathfrak{m} \in \text{Max}(A)} (A_{\mathfrak{m}})^\circ$$

We end up with a diagram similar to our goal:

$$\begin{array}{ccccc} \mathbf{cAff}^{\text{op}} & \xrightarrow{\text{Max}} & \mathbf{Top} & \xrightarrow{U} & \mathbf{Set} \\ \downarrow & & & & \downarrow \\ \mathbf{Alg}^{\text{op}} & \xrightarrow{(-)^\circ} & & & \mathbf{Coalg} \end{array}$$

But it **does not** commute!

Theorem (R., 2024)

Let $F: \mathbf{Alg}^{\text{op}} \rightarrow \mathbf{Coalg}$ be a subfunctor of $(-)^\circ$. If F satisfies $F(\mathbb{M}_d(k)) = \mathbb{M}_d(k)^*$ for all d , then $F = (-)^\circ$.

Fully RFD algebras

What does it mean for A to have “many” f.d. representations?

Def: A is left **fully residually finite-dimensional** if every f.g. left module is a subdirect product of f.d. left modules.

Fully RFD algebras

What does it mean for A to have “many” f.d. representations?

Def: A is left **fully residually finite-dimensional** if every f.g. left module is a subdirect product of f.d. left modules.

Thm: A is left fully RFD $\iff A^\circ$ is a cogenerator in $A\text{-Mod}$.

What algebras satisfy this condition?

- Affine noetherian PI algebras
- In particular, lots of “quantum algebras” at roots of unity

Morita “variance” of A°

In some flavors of NCG, Morita equivalent algebras represent “the same” NC space.

By contrast, functorial spectral theory suggests that Morita equivalent algebras should represent *Morita equivalent spaces*.

Thus I expect there to be Morita equivalence even at the level of noncommutative sets.

Morita “variance” of A°

In some flavors of NCG, Morita equivalent algebras represent “the same” NC space.

By contrast, functorial spectral theory suggests that Morita equivalent algebras should represent *Morita equivalent spaces*.

Thus I expect there to be Morita equivalence even at the level of noncommutative sets.

What this looks like for the finite dual:

Theorem (R., 2024)

If A and B are k -linearly Morita equivalent, then A° and B° are Morita-Takeuchi equivalent.

Understanding A° functorially

Putting functoriality to the test

Problem

If A° is an important invariant, how can we compute it?

Goal: Compute it functorially relative to *subalgebras*, and try to maintain some spectral intuition. Two cases:

- Finite dual of a twisted tensor product
- Relating A° to $Z(A)^\circ$

Twisted tensor products

Many concrete examples of NC algebras arise as **twisted tensor products** of algebras A and B : Given a suitably “nice” linear map

$$\rho: B \otimes A \rightarrow A \otimes B,$$

$A \#_{\rho} B$ is the vector space $A \otimes B$ with multiplication

$$m_{\rho} := (m_A \otimes m_B) \circ (\text{id}_A \otimes \rho \otimes \text{id}_B)$$

Familiar examples:

- Tensor product $A \otimes B$ (if ρ is just the “tensor swap”)
- Ore extensions $A[t; \sigma, \delta] \cong A \#_{\rho} k[t]$
- Smash products $A \# H$, where $H \curvearrowright A$

Crossed product coalgebras

Q: Can we describe $(A \#_{\rho} B)^{\circ}$ in terms of A° and B° ?

For coalgebras C and D and a suitably “nice” linear map

$$\phi: C \otimes D \rightarrow D \otimes C$$

we can form a **crossed product coalgebra** $C \#^{\phi} D$ in a formally dual manner (Caenepeel, Ion, Militaru, Zhu 2000).

Naive guess: Is $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\phi} B^{\circ}$ for some map ϕ ?

Crossed product coalgebras

Q: Can we describe $(A \#_{\rho} B)^{\circ}$ in terms of A° and B° ?

For coalgebras C and D and a suitably “nice” linear map

$$\phi: C \otimes D \rightarrow D \otimes C$$

we can form a **crossed product coalgebra** $C \#^{\phi} D$ in a formally dual manner (Caenepeel, Ion, Militaru, Zhu 2000).

Naive guess: Is $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\phi} B^{\circ}$ for some map ϕ ?

Not always! For $A = k[x]$ and $B = k[y]$, choose ρ so that $A \#_{\rho} B$ has no f.d. modules. Then $(A \# B)^{\circ} = 0 \neq A^{\circ} \#^{\phi} B^{\circ}$ for any ϕ !

Topology to the rescue

Better guess: $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\phi} B^{\circ}$ whenever $\phi = \rho^{\circ}$ makes sense.

This hints that we should interpret $(-)^{\circ}$ as a functor beyond just algebras...

Topology to the rescue

Better guess: $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\phi} B^{\circ}$ whenever $\phi = \rho^{\circ}$ makes sense.

This hints that we should interpret $(-)^{\circ}$ as a functor beyond just algebras...

Denote:

- $\mathbf{Top}_k$ the category of linearly topologized k -vector spaces
- $\mathbf{CF}_k$ full subcategory of spaces with *cofinite* topology (all open subspaces have finite codimension)
- Continuous dual $(-)^{\circ} := \mathbf{Top}_k(-, k): \mathbf{Top}_k^{\text{op}} \rightarrow \mathbf{Vect}_k$

What justifies the notation?

Continuous duality

Monoidal structure: $E \otimes^! F$ is the tensor product topologized by the subspaces $E \otimes F_0 + E_0 \otimes F$ for open $E_0 \subseteq E, F_0 \subseteq F$.

Theorem (R., 2025)

The continuous dual forms a lax monoidal functor

$$(-)^\circ: (\mathbf{Top}_k, \otimes^!, k)^{\text{op}} \rightarrow (\mathbf{Vect}_k, \otimes, k)$$

which restricts to a strong monoidal functor on $(\mathbf{CF}_k, \otimes^!, k)$.

Lax monoidal functor: $E^\circ \otimes F^\circ \hookrightarrow (E \otimes F)^\circ$

Strong monoidal functor: $E^\circ \otimes F^\circ \xrightarrow{\sim} (E \otimes F)^\circ$

The finite dual is a continuous dual

Corollary

Let A be an algebra. Equip it with the linear topology whose open ideals are the ideals of finite codimension. Then the continuous dual of A is the finite dual A° and

$$\Delta_{A^\circ} = m_A^\circ: A^\circ \rightarrow (A \otimes^! A)^\circ \xrightarrow{\sim} A^\circ \otimes A^\circ.$$

Idea: As a vector space,

$$\begin{aligned} \mathbf{Top}_k(A, k) &= \{\phi \in A^* \mid \ker \phi \text{ open}\} \\ &= \{\phi \in A^* \mid \ker \phi \text{ contains fin. codim. } I \trianglelefteq A\} \\ &= A^\circ \end{aligned}$$

Multiplication $m_A: A \otimes^! A \rightarrow A$ is always *continuous* w.r.t. this topology, so we can dualize as in the statement.

Duals of twisted tensor products

Theorem (R., 2025)

Let A and B be algebras with a twisting map ρ . Suppose that ρ is continuous when viewed as a map

$$\rho: B \otimes^! A \rightarrow A \otimes^! B.$$

Then $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\rho^{\circ}} B^{\circ}$.

Duals of twisted tensor products

Theorem (R., 2025)

Let A and B be algebras with a twisting map ρ . Suppose that ρ is continuous when viewed as a map

$$\rho: B \otimes^! A \rightarrow A \otimes^! B.$$

Then $(A \#_{\rho} B)^{\circ} \cong A^{\circ} \#^{\rho^{\circ}} B^{\circ}$.

Fact: ρ continuous $\iff A \otimes^! B \xrightarrow{\sim} A \#_{\rho} B$ in $\mathbf{Top}_k$.

When does this happen?

A sufficient condition: There exist $A_0 \subseteq A$ and $B_0 \subseteq B$ finite extensions such that ρ restricts to the “tensor swap” on $A_0 \otimes B$ and $A \otimes B_0$.

A few applications

Ex: $\theta \in \text{Aut}(A)$ finite order and $\delta \in \text{Der}_\theta(A)$ nilpotent, then

$$A[t; \theta, \delta]^\circ \cong A^\circ \#^\phi k[t]^\circ$$

Ex: A a left Hopf H -module algebra. Suppose A is finite over A^H and $H \curvearrowright A$ factors through a f.d. Hopf algebra. Then

$$(A \# H)^\circ \cong A^\circ \#^\phi H^\circ$$

Ex: Fix bialgebras A and B with twisting and cotwisting maps forming a crossed product bialgebra $H = A \#_\rho^\phi B$. If ρ, ϕ are continuous with respect to $A \otimes^! B$ and $B \otimes^! A$, then as bialgebras

$$H^\circ \cong A^\circ \#_{\phi^\circ}^{\rho^\circ} B^\circ$$

The center and R°

How can we understand R° relative to $\operatorname{Spec}(Z(R))$? Helps to use the **Azumaya locus** (after Brown-Goodearl). Assume $k = \bar{k}$.

Suppose R is prime, affine, and module-finite over $Z(R)$. Let d be the PI-degree of R . Then its *Azumaya locus* is

$$\begin{aligned}\mathcal{A}(R) &= \{\mathfrak{m} = Z(R) \cap \operatorname{Ann}(V) \mid {}_R V \text{ simple, } \dim_k(V) = d\} \\ &= \{\mathfrak{m} \in \operatorname{Max} Z(R) \mid R_{\mathfrak{m}} \text{ is Azumaya over } Z(R)_{\mathfrak{m}}\} \\ &= \{\mathfrak{m} \in \operatorname{Max} Z(R) \mid R\mathfrak{m} \text{ is maximal in } R\} \\ &= \{\mathfrak{m} \in \operatorname{Max} Z(R) \mid R/R\mathfrak{m} \cong \mathbb{M}_d(k)\}.\end{aligned}$$

This is a nonempty open (therefore dense) subset of $\operatorname{Max} Z(R)$.

The Azumaya locus in R°

$$\mathcal{A}(R) = \{\mathfrak{m} \in \text{Max} Z(R) \mid R/R\mathfrak{m} \cong \mathbb{M}_d(k)\}$$

We require a scheme-theoretic version of the Azumaya locus.

The maximal ideals $M = R\mathfrak{m}$ with $\mathfrak{m} \in \mathcal{A}(R)$ are called **regular**.

Denote:

- $N = \bigcap \{M \in \text{Max}(R) \mid M \text{ is not regular}\} \trianglelefteq R$
- $I_R = Z(R) \cap N$ (then $\mathfrak{m} \in \text{Max} Z(R) \setminus \mathcal{A}(R) \iff \mathfrak{m} \supseteq I_R$)

Def: The **Azumaya subscheme** is $U_R = D(I_R) \hookrightarrow \text{Spec} Z(R)$.

It's an open subscheme whose closed points are exactly $\mathcal{A}(R)$.

The Azumaya locus in R°

- $N = \bigcap \{M \in \text{Max}(R) \mid M \text{ is not regular}\} \trianglelefteq R$
- $I_R = Z(R) \cap N$ (then $\mathfrak{m} \in \text{Max} Z(R) \setminus \mathcal{A}(R) \iff \mathfrak{m} \supseteq I_R$)
- $U_R = D(I_R) \hookrightarrow \text{Spec} Z(R)$

Theorem (R., 2024)

If R is prime, affine, and module-finite over $Z(R)$, then

$$R^\circ \cong \varinjlim_{i \rightarrow \infty} (R/N^i)^\circ \oplus (\mathbb{M}^d \otimes \text{Dist}(U_R)).$$

The Azumaya locus in R°

- $N = \bigcap \{M \in \text{Max}(R) \mid M \text{ is not regular}\} \trianglelefteq R$
- $I_R = Z(R) \cap N$ (then $\mathfrak{m} \in \text{Max} Z(R) \setminus \mathcal{A}(R) \iff \mathfrak{m} \supseteq I_R$)
- $U_R = D(I_R) \hookrightarrow \text{Spec} Z(R)$

Theorem (R., 2024)

If R is prime, affine, and module-finite over $Z(R)$, then

$$R^\circ \cong \varinjlim_{i \rightarrow \infty} (R/N^i)^\circ \oplus (\mathbb{M}^d \otimes \text{Dist}(U_R)).$$

What is this saying?

- First summand is a “noncommutative formal neighborhood of non-Azumaya locus”
- Second summand is a “(qudit) $\times$ (Azumaya locus)”

“Picturing” the quantum plane $\mathcal{O}_q(k^2) = k\langle x, y \mid yx = qxy \rangle$

$\mathcal{O}_q(k^2) = k[x] \#_{\rho} k[y]$ where $\rho = \rho_q: y^j \otimes x^i \mapsto q^{ij} x^i \otimes y^j$. This is continuous if q is a root of unity, say of order n . We get

$$\mathcal{O}_q(k^2)^{\circ} \cong k[x]^{\circ} \#^{\rho^{\circ}} k[y]^{\circ} \cong \text{Dist}(\mathbb{A}_k^1) \#^{\phi_q} \text{Dist}(\mathbb{A}_k^1).$$

Assume $k = \bar{k}$, set $G_q = \langle q \rangle^2 \subseteq k^2 = \mathbb{A}_k^2$, which acts on the plane by multiplication. We view $Z(\mathcal{O}_q(k^2)) = k[x^n, y^n]$ as the coordinate ring of the quotient

$$\text{Spec } Z(\mathcal{O}_q(k^2)) \cong \mathbb{A}_k^2 / G_q.$$

Then $N = (xy)$ and the Azumaya subscheme is

$$U_q := D(xy) / G_q \subseteq \mathbb{A}^2 / G_q.$$

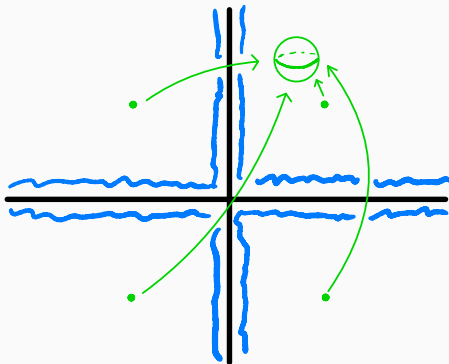
By the Theorem,

$$\mathcal{O}_q(k^2)^{\circ} \cong \varinjlim_i (\mathcal{O}_q(k^2) / (xy)^i)^{\circ} \oplus (\mathbb{M}^n \otimes \text{Dist } U_q)$$

“Picturing” the quantum plane

$$\begin{aligned}\mathcal{O}_q(k^2)^\circ &\cong \text{Dist}(\mathbb{A}_k^1) \#^{\phi_q} \text{Dist}(\mathbb{A}_k^1) \\ &\cong \varinjlim (\mathcal{O}_q(k^2)/(xy)^i)^\circ \oplus (\mathbb{M}^n \otimes \text{Dist } U_q)\end{aligned}$$

Case $n = 2$: $q = -1$ and $\mathbb{M}^2 = \text{qubit}$



Topology/geometry with coalgebras?

Natural next questions to push this perspective further:

(1) Is there a way to equip dual coalgebras with something like a topology and a structure sheaf? Work by [So Nakamura](#) suggests that the answer is yes...

(2) Does the enrichment of algebras over coalgebras allow a way around the big Zariski site obstruction? Recent work on base change of enriched sheaves by [Ariel Rosenfield](#) could help compare a “quantum big Zariski site” with the classical one...

Some questions

1. One could similarly construct an “underlying coalgebra” for noncommutative projective spaces. What are some tools to understand these?
2. The *coradical* of A° describes the simple A -modules, and it deforms nicely in the quantum plane example. How often should we expect it to deform nicely in families $A(q)$?
3. Results of Segal and Le Bruyn suggest that one should be able to reconstruct A° from an A_∞ -structure on the simple A -modules. How to do this in a “Morita variant” way?

Epilogue: where to go from here?

The big question

Question

What should “noncommutative sets” be in general?

$$\begin{array}{ccccc} \mathbf{cRing}^{\text{op}} & \xrightarrow{\text{Spec}} & \mathbf{Top} & \xrightarrow{U} & \mathbf{Set} \\ \downarrow & & & & \downarrow \\ \mathbf{Ring}^{\text{op}} & \xrightarrow{\Sigma} & & & \mathfrak{S} \end{array}$$

It doesn't seem reasonable to construct a “dual coalgebra” spectrum for general rings.

Because, of course, we can't think in terms of duality relative to a field.

The big question

Question

What should “noncommutative sets” be in general?

$$\begin{array}{ccccc} \mathbf{cRing}^{\text{op}} & \xrightarrow{\text{Spec}} & \mathbf{Top} & \xrightarrow{U} & \mathbf{Set} \\ \downarrow & & & & \downarrow \\ \mathbf{Ring}^{\text{op}} & \xrightarrow{\Sigma} & & & \mathfrak{S} \end{array}$$

It doesn't seem reasonable to construct a “dual coalgebra” spectrum for general rings.

Because, of course, we can't think in terms of duality relative to a field. Or can we...?

Making Spec representable

Recall: One motivation for A° came from the isomorphism

$$\mathrm{Max}(A) \cong \mathrm{Hom}(A, k),$$

which let us replace a *representable* functor by an enriched-hom.

Making Spec representable

Recall: One motivation for A° came from the isomorphism

$$\mathrm{Max}(A) \cong \mathrm{Hom}(A, k),$$

which let us replace a *representable* functor by an enriched-hom.

If $R \in \mathbf{cRing}$, there is a sense in which

$$\mathrm{Spec}(R) \cong \mathrm{Hom}(R, \text{fields}),$$

but the functor isn't really representable.

Making Spec representable

Recall: One motivation for A° came from the isomorphism

$$\mathrm{Max}(A) \cong \mathrm{Hom}(A, k),$$

which let us replace a *representable* functor by an enriched-hom.

If $R \in \mathbf{cRing}$, there is a sense in which

$$\mathrm{Spec}(R) \cong \mathrm{Hom}(R, \text{fields}),$$

but the functor isn't really representable.

Amazing fact: The functor Spec is representable if we pass to a larger category of *hyperrings*!

Hypergroups

Def (F. Marty 1934): A **hypergroup** is a set G with a hyperoperation:

$$\star: G \times G \rightarrow \mathcal{P}(G) \setminus \{\emptyset\},$$

that satisfies axioms of

- associativity: $x \star (y \star z) = (x \star y) \star z$ (as subsets);
- identity: exists $e \in G$ such that $x \star e = \{x\} = e \star x$;
- unique inverses: $e \in x^{-1} \star x$ and $e \in x \star x^{-1}$;
- reversibility: $x \in y \star z \implies y \in x \star z^{-1}$ and $z \in y^{-1} \star x$.

Ex: Given a non-normal subgroup K of a group H , double cosets $G = H//K$ form a hypergroup under $xK \star yK = \{zK \mid zK \subseteq xKyK\}$.

Hyperrings

Def (Krasner 1957): A **hyperring** R is both a multiplicative monoid (in the ordinary sense) and an additive reversible hypergroup, satisfying distributivity of the form $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$ in $\mathcal{P}(R)$.

Morphisms: $\phi(xy) = \phi(x)\phi(y)$ and $\phi(x + y) \subseteq \phi(x) + \phi(y)$.

Hyperrings

Def (Krasner 1957): A **hyperring** R is both a multiplicative monoid (in the ordinary sense) and an additive reversible hypergroup, satisfying distributivity of the form $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$ in $\mathcal{P}(R)$.

Morphisms: $\phi(xy) = \phi(x)\phi(y)$ and $\phi(x + y) \subseteq \phi(x) + \phi(y)$.

Example: If R is a ring and $G \leq U(R)$ normalizes R , then R/G becomes a hypergroup under $\bar{x} + \bar{y} = \{\bar{z} \mid zG \subseteq xG + yG\}$.

Key Example: The **Krasner hyperfield** $\mathbf{K} = \{0, 1\}$ has usual multiplication, hyperaddition given by $1 + 1 = \{0, 1\}$. It is the *terminal* hyperfield.

Connes-Consani (2011): For any commutative (hyper)ring R ,

$$\mathrm{Spec}(R) \cong \mathrm{Hom}(R, \mathbf{K})$$

Even more, if X is any scheme, $|X| \cong \mathrm{Hom}(\mathrm{Spec}(\mathbf{K}), X)$!

Connes-Consani (2011): For any commutative (hyper)ring R ,

$$\mathrm{Spec}(R) \cong \mathrm{Hom}(R, \mathbf{K})$$

Even more, if X is any scheme, $|X| \cong \mathrm{Hom}(\mathrm{Spec}(\mathbf{K}), X)$!

Preposterous idea: Perhaps $\mathbf{K}$ is a good “universal base” from which we should study spectral theory??

Connes-Consani (2011): For any commutative (hyper)ring R ,

$$\mathrm{Spec}(R) \cong \mathrm{Hom}(R, \mathbf{K})$$

Even more, if X is any scheme, $|X| \cong \mathrm{Hom}(\mathrm{Spec}(\mathbf{K}), X)$!

Preposterous idea: Perhaps $\mathbf{K}$ is a good “universal base” from which we should study spectral theory??

Q: Does every ring have a “dual $\mathbf{K}$ -coalgebra” $R^\diamond = [R, \mathbf{K}]$?
And if so, is it worthy to be considered a *NC spectrum*?

Still far from an answer but...

Developing hyperalgebra

Test problem (with [So Nakamura](#)): Organize hyperrings and their modules into a category with good properties, then develop a $\mathbf{K}$ -rep. theory for rings...

- Fundamental problem: (canonical) hypergroups don't form a good category! (Neither complete nor cocomplete.)

Developing hyperalgebra

Test problem (with [So Nakamura](#)): Organize hyperrings and their modules into a category with good properties, then develop a $\mathbf{K}$ -rep. theory for rings...

- Fundamental problem: (canonical) hypergroups don't form a good category! (Neither complete nor cocomplete.)
- Source of the problem: $x \star y \neq \emptyset$

Developing hyperalgebra

Test problem (with [So Nakamura](#)): Organize hyperrings and their modules into a category with good properties, then develop a $\mathbf{K}$ -rep. theory for rings...

- Fundamental problem: (canonical) hypergroups don't form a good category! (Neither complete nor cocomplete.)
- Source of the problem: $x \star y \neq \emptyset$
- Solution: allow empty products, which means dropping associativity!

Developing hyperalgebra

Test problem (with [So Nakamura](#)): Organize hyperrings and their modules into a category with good properties, then develop a **K**-rep. theory for rings...

- Fundamental problem: (canonical) hypergroups don't form a good category! (Neither complete nor cocomplete.)
- Source of the problem: $x \star y \neq \emptyset$
- Solution: allow empty products, which means dropping associativity!
- New category **Msc** of **mosaics**: $(M, \star, e)$ has identity, reversibility, not necessarily associative.

Benefits of nonassociative hyperalgebra

Some results proved in [Nakamura-R., 2025]:

- **Msc** and **cMsc** are complete, cocomplete, and regular categories.
- There is a tensor product on **cMsc** that gives a tensor-hom adjunction, i.e. closed monoidal category $(\mathbf{cMsc}, \otimes, \mathbb{F})$
- Hyperrings embed fully faithfully into the category of monoid objects **Mon(cMsc)**.

Benefits of nonassociative hyperalgebra

Some results proved in [Nakamura-R., 2025]:

- **Msc** and **cMsc** are complete, cocomplete, and regular categories.
- There is a tensor product on **cMsc** that gives a tensor-hom adjunction, i.e. closed monoidal category $(\mathbf{cMsc}, \otimes, \mathbb{F})$
- Hyperrings embed fully faithfully into the category of monoid objects $\mathbf{Mon}(\mathbf{cMsc})$.

Thus monoid objects in **cMsc** generalize hyperrings...

Next steps

Work-in-progress [Nakamura-R.]...

Def: The category of **hoops** is $\mathbf{Hoop} = \mathbf{Mon}(\mathbf{cMsc})$.

Some highlights thus far:

- **Hoop** is complete, cocomplete, and regular (basically because **cMsc** is), as well as **Mod- H** for each hoop H .
- We can form tensor products of (bi)modules over hoops, with tensor-hom adjunction
- The “field-like” objects of **Hoop** already appeared by another name: **pastures** defined by Baker-Lorscheid.
- **K** is terminal among all “division-ring-like” hoops.

Rudimentary representation theory over K

We define $M \in \mathbf{Mod}\text{-}H$ to be **very simple** if there are no nontrivial regular epimorphisms out of M .

Degenerate Schur's Lemma: If M_H is very simple, then $\text{End}_H(M) = \{0, \text{id}_M\} \cong \mathbb{F}_2$ or K .

Includes the case where $H = R$ is a ring...

Rudimentary representation theory over K

We define $M \in \mathbf{Mod}\text{-}H$ to be **very simple** if there are no nontrivial regular epimorphisms out of M .

Degenerate Schur's Lemma: If M_H is very simple, then $\text{End}_H(M) = \{0, \text{id}_M\} \cong \mathbb{F}_2$ or K .

Includes the case where $H = R$ is a ring...

Commutative case: For every commutative ring R ,

$$\begin{aligned}\text{Spec}(R) &\cong \{\text{very simple modules over } R\} \\ &\cong \{\text{very simple modules over } R \boxtimes K\}\end{aligned}$$

The map is $\mathfrak{p} \mapsto Q(R/\mathfrak{p}) \boxtimes_{Q(R/\mathfrak{p})} K$

Rudimentary representation theory over K

This extends Connes-Consani in a satisfying way: the correspondences for affine algebras over $k = \bar{k}$

$$\mathrm{Hom}(A, k) \overset{\sim}{\longleftarrow} \mathrm{Max}(A) \overset{\sim}{\longrightarrow} \{\text{irreps of } A \text{ over } k\}$$

Rudimentary representation theory over K

This extends Connes-Consani in a satisfying way: the correspondences for affine algebras over $k = \bar{k}$

$$\mathrm{Hom}(A, k) \overset{\sim}{\longleftarrow} \mathrm{Max}(A) \overset{\sim}{\longrightarrow} \{\text{irreps of } A \text{ over } k\}$$

now becomes the following for commutative rings:

$$\mathrm{Hom}(R, K) \overset{\sim}{\longleftarrow} \mathrm{Spec}(R) \overset{\sim}{\longrightarrow} \{\text{irreps of } R \text{ over } K\}$$

Rudimentary representation theory over K

This extends Connes-Consani in a satisfying way: the correspondences for affine algebras over $k = \bar{k}$

$$\mathrm{Hom}(A, k) \xleftarrow{\sim} \mathrm{Max}(A) \xrightarrow{\sim} \{\text{irreps of } A \text{ over } k\}$$

now becomes the following for commutative rings:

$$\mathrm{Hom}(R, K) \xleftarrow{\sim} \mathrm{Spec}(R) \xrightarrow{\sim} \{\text{irreps of } R \text{ over } K\}$$

Not-so-preposterous idea? Perhaps K is a good “universal base” from which we should study spectral theory.

Representation theory over K

$$\mathrm{Hom}(R, K) \overset{\sim}{\longleftarrow} \mathrm{Spec}(R) \overset{\sim}{\longrightarrow} \{\text{irreps of } R \text{ over } K\}$$

How could we describe “ K -irreps” of a noncommutative ring?

Representation theory over K

$$\mathrm{Hom}(R, K) \overset{\sim}{\longleftarrow} \mathrm{Spec}(R) \overset{\sim}{\longrightarrow} \{\text{irreps of } R \text{ over } K\}$$

How could we describe “ K -irreps” of a noncommutative ring?

There is a way to convert (ordinary) simple R -modules into very simple $K \boxtimes R$ -modules:

$$\begin{array}{l} V_R \text{ simple,} \\ D = \mathrm{End}_R(V) \end{array} \rightsquigarrow K \boxtimes_D V \text{ very simple}$$

Remaining problem: Can we characterize *all* very simple right $K \boxtimes R$ -modules in terms of, say, “prime right ideals”?

Thank you!